

$$3(1) \begin{bmatrix} 2 & 2 & -4 & 3 & 0 \\ 2 & 4 & -11 & 10 & 0 \\ 4 & 2 & -1 & 8 & 0 \\ 1 & 1 & -3 & 3 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_{14} \rightarrow \\ R_2 - 2R_1 \\ R_4 - 2R_1 \\ R_3 - 4R_1 \end{array} \begin{bmatrix} 1 & 1 & -3 & 3 & 0 \\ 0 & 2 & -5 & 4 & 0 \\ 0 & -2 & 11 & -4 & 0 \\ 0 & 0 & 2 & -3 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_1 + R_2 \\ CR_2 \\ R_4 - 2R_2 \\ CR_4 \\ CR_3 \end{array} \begin{bmatrix} 1 & 1 & -3 & 3 & 0 \\ 0 & 1 & -5/2 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$x_1 = x_2 = x_3 = x_4 = 0$$

$$10(4) \begin{bmatrix} 1 & 2 & 4 & 2 & 2 \\ 3 & 5 & 10 & 7 & 5 \\ 1 & -3 & -9 & 9 & 1 \\ 2 & -3 & -6 & 14 & b \end{bmatrix}$$

$$\begin{array}{l} R_2 - 3R_1 \\ R_3 - R_1 \\ R_4 - 2R_1 \\ -R_2 \end{array} \begin{bmatrix} 1 & 2 & 4 & 2 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & -5 & -4 & 7 & -1 \\ 0 & -7 & -14 & 10 & b-4 \end{bmatrix}$$

$$\begin{array}{l} R_1 + 5R_2 \\ R_4 + 7R_2 \end{array} \begin{bmatrix} 1 & 2 & 4 & 2 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 6 & 9 & 4 \\ 0 & 0 & 0 & 3 & b+3 \end{bmatrix}$$

i) $a = b \quad b \neq 3$

no solution

ii) $a = b \quad b = 3$

$$\begin{cases} x_1 = -8 \\ x_2 = 3-2a \\ x_3 = c \\ x_4 = 2 \end{cases} \text{ infinite number of solutions}$$

iii) $a \neq b$

$$\begin{array}{l} CR_3, CR_4 \end{array} \begin{bmatrix} 1 & 2 & 4 & 2 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & \frac{2}{b-a} & \frac{4}{b-a} \\ 0 & 0 & 0 & 1 & \frac{b+3}{3} \end{bmatrix}$$

$$\begin{cases} x_1 = \frac{-4b-12}{3} \\ x_2 = \frac{24-ab+12b-6a}{18-3a} \\ x_3 = \frac{b-2b}{18-3a} \\ x_4 = \frac{b+3}{3} \end{cases}$$

5 $A = \begin{bmatrix} 1 & 0 & 1 & 2 & 3 \\ 0 & 1 & 4 & 5 & 6 \\ \hline 1 & 2 & 1 & 2 & 3 \\ 4 & 5 & 4 & 5 & 6 \\ 7 & 8 & 7 & 8 & 9 \end{bmatrix}$

$$11 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 2 & a & 0 \\ 1 & 4 & a^2 & 0 \\ 1 & 2 & 1 & a-1 \end{array} \right]$$

$$\begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & a-1 & 0 \\ 0 & 2 & a^2-1 & 0 \\ 0 & 1 & 0 & a-1 \end{array} \right]$$

$$\begin{array}{l} R_3 - 2R_2 \\ R_4 - R_2 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & a-1 & 0 \\ 0 & 0 & (a-1)^2 & 0 \\ 0 & 0 & 1-a & a-1 \end{array} \right]$$

i) $a = 1$

$$x_1 = x_2 = x_3 = 0$$

ii) $a \neq 1$

no solutions

$$12 \text{ pf. } \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & a_1 \\ 0 & 1 & -1 & 0 & 0 & a_2 \\ 0 & 0 & 1 & -1 & 0 & a_3 \\ 0 & 0 & 0 & 1 & -1 & a_4 \\ -1 & 0 & 0 & 0 & 1 & a_5 \end{array} \right]$$

$$\xrightarrow{R_5 + R_2 + R_3 + R_4} \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & a_1 \\ 0 & 1 & -1 & 0 & 0 & a_2 \\ 0 & 0 & 1 & -1 & 0 & a_3 \\ 0 & 0 & 0 & 1 & -1 & a_4 \\ 0 & 0 & 0 & 0 & 0 & \sum a_i \end{array} \right]$$

if equations have a solution.

$$\sum_{i=1}^5 0x_i = 0 = a_1 + a_2 + a_3 + a_4 + a_5$$

② if $a_1 + a_2 + a_3 + a_4 + a_5 = 0$

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & a_1 \\ 0 & 1 & -1 & 0 & 0 & a_2 \\ 0 & 0 & 1 & -1 & 0 & a_3 \\ 0 & 0 & 0 & 1 & -1 & a_4 \\ 0 & 0 & 0 & 0 & 0 & \sum a_i \end{array} \right]$$

$$\begin{array}{l} R_3 + R_4 \\ R_2 + R_3 \\ R_1 + R_2 \end{array} \rightarrow \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -1 & a_1 + a_2 + a_3 + a_4 \\ 0 & 1 & 0 & 0 & -1 & a_2 + a_3 + a_4 \\ 0 & 0 & 1 & 0 & -1 & a_3 + a_4 \\ 0 & 0 & 0 & 1 & -1 & a_4 \\ 0 & 0 & 0 & 0 & 0 & \sum a_i \end{array} \right]$$

$$\left\{ \begin{array}{l} x_1 = -a_5 \\ x_2 = -a_1 - a_5 \\ x_3 = -a_2 - a_4 \\ x_4 = -a_3 \\ x_5 = -a_1 \end{array} \right.$$

习题 2

$$1 (1) \tau(135786492) = 13, \text{ odd}$$

$$(3) \tau[135 \cdots (2n-1)(2n)(2n-2)(2n-4) \cdots 2]$$

$$= 2 + 4 + 6 + \cdots + (2n-2)$$

$$= n^2 - n, \text{ even}$$

$$5 (1) \tau(2145367) = 3$$

$$\text{sign} \rightarrow (-1)^3 = -1$$

$$(2) \tau(2357614) = 1$$

$$\tau(5412673) = 10$$

$$\text{sign} \rightarrow (-1)^{1+10} = -1$$

$$6 (2) \begin{vmatrix} a_1 & a_2 & a_3 & a_4 & a_5 \\ b_1 & b_2 & b_3 & b_4 & b_5 \\ c_1 & c_2 & 0 & 0 & 0 \\ d_1 & d_2 & 0 & 0 & 0 \\ e_1 & e_2 & 0 & 0 & 0 \end{vmatrix} = \sum_{(j_1, \dots, j_5)} (-1)^{\tau(j_1, \dots, j_5)} a_{1j_1} \cdots a_{5j_5}$$

$$= a_1 b_2 \cdot 0 \cdot 0 \cdot 0 + b_1 a_2 \cdot 0 \cdot 0 \cdot 0 + \cdots$$

$$= 0$$

every product has 5 elements taken from every column and row

by the Pigeonhole Principle, it's impossible to take 3 non-zero element from two different columns.

$$\Rightarrow D = 0$$

$$8 (1) \begin{vmatrix} 2 & -2 & -4 & 0 \\ -1 & 4 & 3 & 5 \\ 1 & 3 & -2 & -3 \\ 0 & 2 & 5 & 1 \end{vmatrix} = - \begin{vmatrix} 1 & 3 & -2 & -3 \\ -1 & 4 & 3 & 5 \\ 2 & -2 & -4 & 0 \\ 0 & 2 & 5 & 1 \end{vmatrix} = -15 \begin{vmatrix} 1 & 3 & -2 & -3 \\ 0 & 2 & 5 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -\frac{1}{2} \end{vmatrix} = -15 \cdot 1 \cdot 2 \cdot 2 \cdot (-\frac{1}{2}) = 270$$

$$\begin{aligned}
 (b) \quad & \begin{vmatrix} a_1 & -a_1 & 0 & \cdots & 0 & 0 \\ 0 & a_2 & -a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & a_n & -a_n \\ b & b & b & \cdots & b & b \end{vmatrix} = \begin{vmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ 1 & 1 & 1 & \cdots & 1 & 1 \end{vmatrix} \cdot b a_1 a_2 \cdots a_n \\
 & = \begin{vmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ n & 1 & 1 & \cdots & 1 & 1 \end{vmatrix} b a_1 a_2 \cdots a_n = (-1)^{n+1} \cdot n \cdot \begin{vmatrix} -1 & 0 & \cdots & 0 & 0 \\ 1 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -1 \end{vmatrix} b a_1 \cdots a_n \\
 & = (-1)^{n+1} \cdot n \cdot (-1)^{n-1} b a_1 \cdots a_n \\
 & = n b a_1 a_2 \cdots a_n
 \end{aligned}$$

$$9 (2) \quad \begin{vmatrix} a_1 - b_1 & a_1 - b_2 & \cdots & a_1 - b_n \\ a_2 - b_1 & a_2 - b_2 & \cdots & a_2 - b_n \\ \vdots & \vdots & & \vdots \\ a_n - b_1 & a_n - b_2 & \cdots & a_n - b_n \end{vmatrix} = 0$$

$$\text{pf} \quad \begin{vmatrix} a_1 & a_1 & \cdots & a_1 \\ a_2 & a_2 & \cdots & a_2 \\ \vdots & \vdots & & \vdots \\ a_n & a_n & \cdots & a_n \end{vmatrix} - \begin{vmatrix} b_1 & b_1 & \cdots & b_1 \\ b_2 & b_2 & \cdots & b_2 \\ \vdots & \vdots & & \vdots \\ b_n & b_n & \cdots & b_n \end{vmatrix} = 0$$

Same column Same row

$$12 (5) \quad \begin{vmatrix} a & b & \cdots & b & b \\ b & a & \cdots & b & b \\ \vdots & \vdots & & \vdots & \vdots \\ b & b & \cdots & a & b \\ b & b & \cdots & b & a \end{vmatrix}_n = \begin{vmatrix} a+(n-1)b & a+(n-1)b & \cdots & a+(n-1)b & a+(n-1)b \\ b & a & \cdots & b & b \\ \vdots & \vdots & & \vdots & \vdots \\ b & b & \cdots & a & b \\ b & b & \cdots & b & a \end{vmatrix} = [a+(n-1)b] \cdot \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ 0 & a-b & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & a-b & 0 \\ 0 & 0 & \cdots & 0 & a-b \end{vmatrix} = [a+(n-1)b] (a-b)^{n-1}$$

习题 2

$$\begin{aligned}
 12 \quad (2) \quad & \begin{vmatrix} \lambda & -1 & 0 & 0 \\ 0 & \lambda & -1 & 0 \\ 0 & 0 & \lambda & -1 \\ 4 & 3 & 2 & \lambda+1 \end{vmatrix} = \begin{vmatrix} \lambda & -1 & 0 & 0 \\ 0 & \lambda & -1 & 0 \\ 0 & 0 & \lambda & -1 \\ 0 & 0 & 0 & \lambda \end{vmatrix} + \begin{vmatrix} \lambda & -1 & 0 & 0 \\ 0 & \lambda & -1 & 0 \\ 0 & 0 & \lambda & -1 \\ 4 & 3 & 2 & 1 \end{vmatrix} \\
 &= \lambda^4 + (-1)(-1)^{3+4} \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 4 & 3 & 2 \end{vmatrix} + 1 \cdot (-1)^{4+4} \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 0 & 0 & \lambda \end{vmatrix} \\
 &= \lambda^4 + \lambda(2\lambda+3) + 4 \cdot 1 + \lambda^3 \\
 &= \lambda^4 + \lambda^3 + 2\lambda^2 + 3\lambda + 4
 \end{aligned}$$

$$(4) \quad D_n = \begin{vmatrix} x & -1 & 0 & \cdots & 0 & 0 \\ 0 & x & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & a_1+x \end{vmatrix} = x(-1)^{1+n} D_{n-1} + a_n(-1)^{1+n}(-1)^{n-1}$$

$$\Rightarrow D_n = x D_{n-1} + a_n$$

$$= x(x D_{n-2} + a_{n-1}) + a_n$$

$$= x^2 D_{n-1} + x a_{n-1} + a_n$$

$$= x^2(x D_{n-3} + a_{n-2}) + x a_{n-1} + a_n$$

$$= \dots$$

$$= x^{n-1} D_1 + \dots + x^2 a_{n-2} + x a_{n-1} + a_n$$

$$= x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-2} x^2 + a_{n-1} x + a_n$$

$$(11) \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 2 & 2^2 & 2^3 & \cdots & 2^n \\ 3 & 3^2 & 3^3 & \cdots & 3^n \\ \vdots & \vdots & \vdots & & \vdots \\ n & n^2 & n^3 & \cdots & n^n \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & \cdots & n \\ 1 & 2^2 & 3^2 & \cdots & n^2 \\ 1 & 2^3 & 3^3 & \cdots & n^3 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 2^n & 3^n & \cdots & n^n \end{vmatrix} = n! \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 3 & \cdots & n \\ 1 & 2^2 & 3^2 & \cdots & n^2 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 2^{n-1} & 3^{n-1} & \cdots & n^{n-1} \end{vmatrix}$$

$$= n! \prod_{1 \leq i < j \leq n} (j - i)$$

$$= n! \cdot 1! \cdot 2! \cdot 3! \cdots (n-1)!$$

$$13) i) D_n = \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 1 & 1 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & 0 & 0 & \cdots & 1 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 1 \end{vmatrix} = 1(-1)^{n+n} D_{n-1} + 1(-1)^{n+1} \cdot n(-1)^{n+1} \cdot 1$$

$$= D_{n-1} - n$$

$$D_n - D_1 = -(2+3+\cdots+n) = -\frac{(n+2)(n-1)}{2}$$

$$D_n = 1 - \frac{n^2+n-2}{2} = 2 - \frac{n(n+1)}{2}$$

$$ii) t_1 A_{11} + \cdots + t_n A_{1n}$$

$$= \begin{vmatrix} t_1 & t_2 & \cdots & t_n \\ 1 & 1 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 1 \end{vmatrix} = t_1 - (t_2 + t_3 + \cdots + t_n)$$

$$14 \quad D_n = \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ x_1 & x_2 & \cdots & x_{n-1} & x_n \\ \vdots & \vdots & & \vdots & \vdots \\ x_1^{n-2} & x_2^{n-2} & \cdots & x_{n-1}^{n-2} & x_n^{n-2} \\ x_1^n & x_2^n & \cdots & x_{n-1}^n & x_n^n \end{vmatrix}_n = \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ x_1 - x_n & x_2 - x_n & \cdots & x_{n-1} - x_n & 0 \\ x_1^2 - x_n x_1 & x_2^2 - x_n x_2 & \cdots & x_{n-1}^2 - x_n x_{n-1} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ x_1^{n-2} - x_n x_1^{n-3} & x_2^{n-2} - x_n x_2^{n-3} & \cdots & x_{n-1}^{n-2} - x_n x_{n-1}^{n-3} & 0 \\ x_1^n - x_n^{n-2} x_1^2 & x_2^n - x_n^{n-2} x_2^2 & \cdots & x_{n-1}^n - x_n^{n-2} x_{n-1}^2 & 0 \end{vmatrix}$$

$$= (-1)^{n+1} (x_1 - x_n) \cdots (x_{n-1} - x_n) \begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_{n-1} \\ \vdots & \vdots & & \vdots \\ x_1^{n-3} & x_2^{n-3} & \cdots & x_{n-1}^{n-3} \\ x_1^{n-2}(x_1+x_n) & x_2^{n-2}(x_2+x_n) & \cdots & x_{n-1}^{n-2}(x_{n-1}+x_n) \end{vmatrix}$$

$$= (x_n - x_1) \cdots (x_n - x_{n-1}) \left[\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_{n-1} \\ \vdots & \vdots & & \vdots \\ x_1^{n-4} & x_2^{n-4} & \cdots & x_{n-1}^{n-4} \end{vmatrix} + x_n \begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_{n-1} \\ \vdots & \vdots & & \vdots \\ x_1^{n-3} & x_2^{n-3} & \cdots & x_{n-1}^{n-3} \end{vmatrix} \right]$$

$$= (x_n - x_1) \cdots (x_n - x_{n-1}) (D_{n-1} + x_n \prod_{1 \leq i < j \leq n-1} (x_j - x_i))$$

$$D_n = (x_n - x_1) \cdots (x_n - x_{n-1}) D_{n-1} + x_n \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

$$= (x_n - x_1) \cdots (x_n - x_{n-1}) [(x_{n-1} - x_1) \cdots (x_{n-1} - x_{n-2}) D_{n-2} + x_{n-1} \prod_{1 \leq i < j \leq n-1} (x_j - x_i)] + x_n \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

$$= (x_n - x_1) \cdots (x_{n-2} - x_{n-2}) D_{n-2} + (x_{n-1} + x_n) \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

$$= \dots$$

$$= (x_1 + x_2 + \cdots + x_n) \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

15 (1)

$$\begin{bmatrix} 1 & 2 & -1 & 3 & | & 2 \\ 2 & -1 & 3 & -2 & | & 7 \\ 0 & 3 & -1 & 1 & | & 6 \\ 1 & -1 & 1 & 4 & | & -4 \end{bmatrix}$$

$$D = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & -2 \\ 0 & 3 & -1 & 1 \\ 1 & -1 & 1 & 4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -8 \\ 0 & 3 & -1 & 1 \\ 0 & -3 & 2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -5 & 5 & -8 \\ 3 & -1 & 1 \\ -3 & 2 & 1 \end{vmatrix} = -39$$

$$D_1 = \begin{vmatrix} 2 & 2 & -1 & 3 \\ 7 & -1 & 3 & -2 \\ 6 & 3 & -1 & 1 \\ -4 & -1 & 1 & 4 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 2 & -1 & 3 \\ 13 & 5 & 0 & 7 \\ 4 & 1 & 0 & -2 \\ -2 & 1 & 0 & 7 \end{vmatrix}$$

$$= (-1)(-1)^{3+1} \begin{vmatrix} 13 & 5 & 7 \\ 4 & 4 & -2 \\ -2 & 1 & 7 \end{vmatrix}$$

$$= -39$$

$$D_2 = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & 7 & 3 & -2 \\ 0 & 6 & -1 & 1 \\ 1 & -4 & 1 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & 3 & 5 & -8 \\ 0 & 6 & -1 & 1 \\ 0 & -6 & 2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 5 & -8 \\ 6 & -1 & 1 \\ -6 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 5 & -8 \\ 0 & -11 & 17 \\ 0 & 12 & -15 \end{vmatrix} = -117$$

$$D_3 = \begin{vmatrix} 1 & 2 & 2 & 3 \\ 2 & -1 & 7 & -2 \\ 0 & 3 & 6 & 1 \\ 1 & -1 & -4 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 2 & 3 \\ 0 & -5 & 3 & -8 \\ 0 & 3 & 6 & 1 \\ 0 & -3 & -6 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -5 & 3 & -8 \\ 3 & 6 & 1 \\ -3 & -6 & 1 \end{vmatrix} = \begin{vmatrix} -5 & 3 & -8 \\ 13 & 0 & 17 \\ -13 & 0 & -15 \end{vmatrix} = -78$$

$$D_4 = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & -2 \\ 0 & 3 & -1 & 1 \\ 1 & -1 & 1 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -8 \\ 0 & 3 & -1 & 1 \\ 0 & -3 & 2 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -5 & 5 & -8 \\ 3 & -1 & 1 \\ -3 & 2 & 1 \end{vmatrix} = -79$$

$$\begin{cases} x_1 = \frac{D_1}{D} = 1 \\ x_2 = \frac{D_2}{D} = 3 \\ x_3 = \frac{D_3}{D} = 2 \\ x_4 = \frac{D_4}{D} = -1 \end{cases}$$

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2026.9.22 W2 Homework

$$16 (3) \begin{vmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ e & 2 & 0 & 0 & 0 & 0 \\ f & g & 3 & 0 & 0 & 0 \\ b_{11} & b_{12} & b_{13} & 0 & 0 & 1 \\ b_{21} & b_{22} & b_{23} & 0 & 2 & u \\ b_{31} & b_{32} & b_{33} & 3 & v & t \end{vmatrix}$$

$$\xrightarrow{\text{Col } 4, 5, 6} (-1)^{1+2+6+4+5+6} \begin{vmatrix} 1 \\ 2 & u \\ 3 & v & t \end{vmatrix} \cdot \begin{vmatrix} 1 \\ e & 2 \\ f & g & 3 \end{vmatrix} = 36$$

$$17 (1) \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & -2 & 4 & 5 \\ 1 & 10 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R.E.} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -4 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank} = 2$$

$$(4) \begin{bmatrix} 1 & 1 & 2 & -2 \\ 1 & 3 & -k & -2k \\ 1 & -1 & 6 & 0 \end{bmatrix}$$

$$\xrightarrow{G.E.} \begin{bmatrix} 1 & 1 & 2 & -2 \\ 0 & 2 & -2-k & -2-2k \\ 0 & 0 & 2-k & 4-2k \end{bmatrix}$$

$$\text{rank} = \begin{cases} 2, & k=2 \\ 3, & k \neq 2 \end{cases}$$